MECH 325
Flexible Drive Elements
Wire Rope



Wire Rope

- Wire rope is used for numerous activities such as hoisting, slinging, securing, reinforcing, etc.
- Wire rope is considered a slow speed flexible drive.
- *Rigging* is a general term used for applications of wire rope.



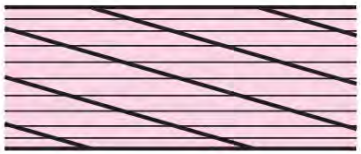


Wire Rope

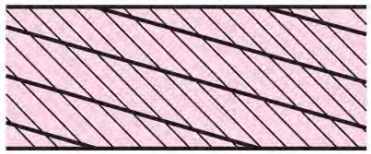
- The spirals or helical twist in the wires and strands are referred to as the lay. It can be considered the pitch of the wire.



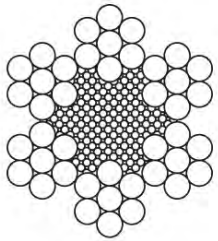
- *Regular Lay* has wires laid opposite to the strands.
- *Lang Lay* strands laid in the same direction as the strand giving higher flexibility and abrasion resistance.



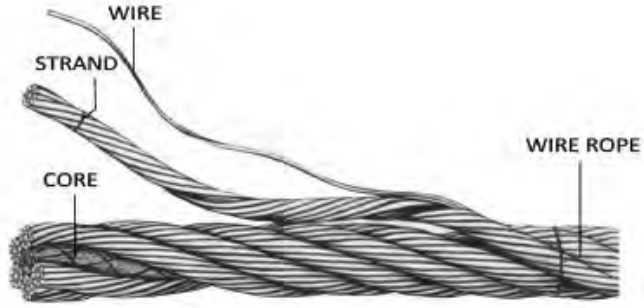
(*a*) Regular lay

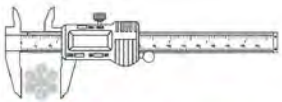


(*b*) Lang lay

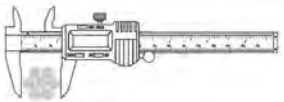


(*c*) Section of 6 × 7 rope





✖ Incorrect

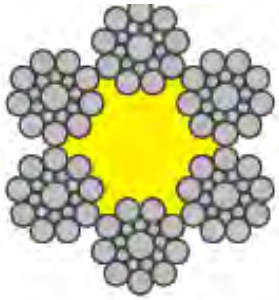


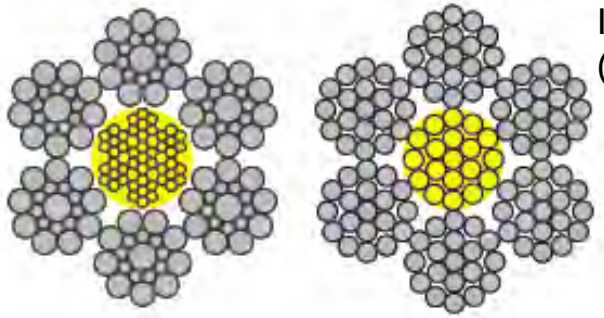
√ Correct

Wire Rope

- Wires are cold drawn steel that are twisted together to form the strand.
- The strands are helically laid together around the core.
- The numerical designation has number of strands and number of wires in each strand. The strand numbers are then defined in brackets by their arraignment in the strand.
- The core can be a soft material like hemp fibre (HC FC), or polyethene (PC) that assists in stabilizing the strands and provides lubricant.
- Steel or wire core are also used to improve strength and avoid crushing on a drum

Fiber Core





Independent wire rope core (IWRC)

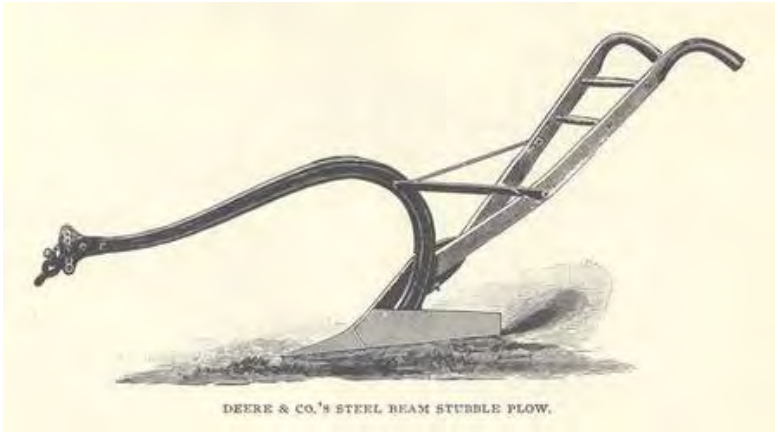
Wire rope

- The grade of wire rope affects the ultimate strength of the rope.
- Wire ropes are specified as Bright (most common), Galvanized, or Stainless Steel.
- Just like drive chains, wire rope must be lubricated when passing over drum or sheaves. This assists the relative sliding motion of the wires and strands as they pass through sheaves and drums.
- All safety procedures must be followed when using wire ropes



Wire Rope

- Wire Rope specifications are normally selected from data supplied by manufactures.
- Ratings are normally specified as *minimum breaking strengths*.
- Steel for the wire was originally compared with the strength of the steel used to make ploughs.

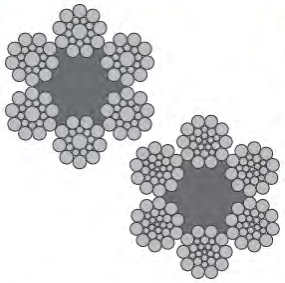


IPS – Improved Plow Steel
XPS – Extra Improved Plow Steel

TABLE 7–25 6×19 and 6×36 Classes Technical Data

	Fiber core			IWRC		
Diameter	Weight per foot	Min breaking force		Weight per foot	Min breaking force	
		IPS	XIP		IPS	XIP
in	lb/ft	tons	tons	lb/ft	tons	tons
1/4	0.105	2.74	3.02	0.116	2.94	3.4
5/16	0.164	4.26	4.69	0.18	4.58	5.27
3/8	0.236	6.1	6.72	0.26	6.56	7.55
7/16	0.32	8.27	9.1	0.35	8.89	10.2
1/2	0.42	10.7	11.8	0.46	11.5	13.3
9/16	0.53	13.5	14.9	0.59	14.5	16.8
5/8	0.66	16.7	18.3	0.72	17.9	20.6
3/4	0.95	23.8	26.2	1.04	25.6	29.4
7/8	1.29	32.2	35.4	1.42	34.6	39.8
1	1.68	41.8	46	1.85	44.9	51.7
1 1/8	2.13	52.6	57.8	2.34	56.5	65
1 1/4	2.63	64.6	71.1	2.89	69.4	79.9
1 3/8	3.18	77.7	85.5	3.5	83.5	96
1 1/2	3.78	92	101	4.16	98.9	114
1 5/8	4.44	107	118	4.88	115	132
1 3/4	5.15	124	137	5.67	133	153
1 7/8	5.91	141	156	6.5	152	174
2	6.72	160	176	7.39	172	198
2 1/8	7.59	179	197	8.35	192	221
2 1/4	8.51	200	220	9.36	215	247

1 ton = 2000 lbf



6 x 19 FC Class Wire Rope

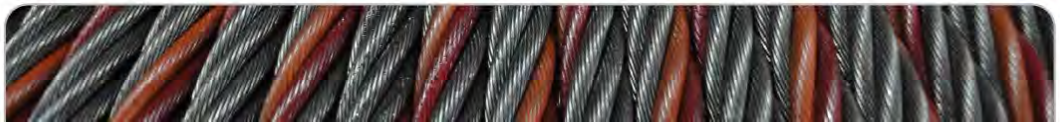
6 Outer Strands, Rotation Wire Rope

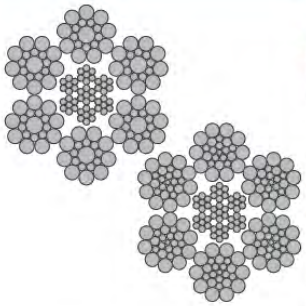
Includes 6x19S & 6x26WS construction types. A good general purpose wire rope which is wear resistant and suitable for sling ropes, marine rope, lashing, fishing, anchor lines and logging.
6x19S (9+9+1) FC, Load bearing outer wires 114
6x26WS (10+(5+5)+5+1) FC, Load bearing outer wires 156
Fill factor 0.49
Use of swivel is not permitted.

Diameter Inches	Weight per Foot Pounds	Grade Tensile	Bright		Galvanized	
			Wesco Code	Minimum Strength Pounds	Wesco Code	Minimum Strength Pounds
1/4	0.11	EIPS	W61908C1	6,040	W61908D1	5,436
5/16	0.16	EIPS	W61910C1	9,380	W61910D1	8,442
3/8	0.24	EIPS	W61912C1	13,420	W61912D1	12,078
7/16	0.32	EIPS	W61914C1	18,180	W61914D1	16,362
1/2	0.42	EIPS	W61916C1	23,600	W61916D1	21,240
9/16	0.53	EIPS	W62618C1	29,800	W61918D1	26,820
5/8	0.66	EIPS	W62620C1	36,600	W61920D1	32,940
3/4	0.95	EIPS	W62624C1	52,400	W61924D1	47,160
7/8	1.29	EIPS	W62628C1	70,800	W61928D1	63,720
1	1.68	EIPS	W62632C1	92,000	W61932D1	82,800
1-1/8	2.13	EIPS	W62636C1	115,800	W61936D1	104,220
1-1/4	2.63	EIPS	W62640C1	142,000	W61940D1	127,800
1-1/4	2.63	EEIPS	~	~	W62640D1C-EEIPS	156,200
1-3/8	3.18	EIPS	W62644C1	170,800	W61944D1	153,720
1-1/2	3.78	EIPS	W62648C1	202,000	W61948D1	181,800
1-5/8	4.4	EIPS	W62652C1	236,000	W61952D1	212,400
1-3/4	5.15	EIPS	W62656C1	272,000	W61956D1	244,800
1-7/8	5.91	EIPS	W62660C1	310,000	W61960D1	279,000
2	6.72	EIPS	W62664C1	352,000	W61964D1	316,800
2-1/4	8.51	EIPS	W62672C1	440,000	W61972D1	396,000
2-1/2	10.50	EIPS	W62680C1	538,000	W62680D1	484,200

EIPS – Extra Improved Plow Steel

EEIPS – Extra, Extra, Improved Plow Steel





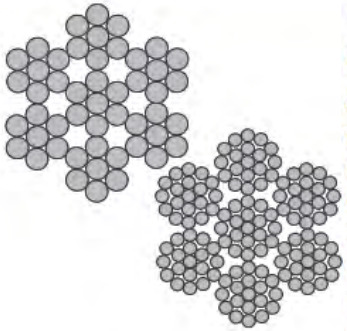
6 x 19 IWRC Class Wire Rope

6 Outer Strands, Rotation Wire Rope

Includes 6x19S & 6x26WS construction types. A good general purpose wire rope which is wear and crush resistant but not very flexible. Suitable for lifting slings (chokers), winch lines, boom pendants, boom hoist ropes, suspension cables, fishing, gang panels and logging.
6x19S (9+9+1) IWRC, Load bearing outer wires 114
6x26WS (10+(5+5)+5+1) IWRC, Load bearing outer wires 156
Fill factor 0.57
Use of swivel is not permitted.

Diameter Inches	Weight per Foot Pounds	Grade Tensile	Bright		Galvanized	
			Wesco Code	Minimum Strength Pounds	Wesco Code	Minimum Strength Pounds
1/4	0.12	EIPS	W61908A1	6,880	W61908B1	6,190
5/16	0.18	EIPS	W61910A1	10,540	W61910B1	9,480
3/8	0.26	EIPS	W61912A1	15,100	W61912B1	13,590
7/16	0.35	EIPS	W61614A1	20,400	W61914B1	18,360
7/16	0.35	EEIPS	~	~	W61914B1-EEIPS	22,400
1/2	0.46	EIPS	W62616A1	26,600	W61916B1	24,000
9/16	0.59	EIPS	W62618A1	33,600	W61918B1	30,240
5/8	0.72	EIPS	W62620A1	41,200	W61920B1	37,080
11/16	0.85	EIPS	W62622A1	54,800	W61922B1	49,394
3/4	1.04	EIPS	W62624A1	58,800	W61924B1	52,920
13/16	1.19	EIPS	W62626A1	76,100	W61926B1	68,554
7/8	1.42	EIPS	W62628A1	79,600	W61928B1	71,640
15/16	1.59	EIPS	W62630A1	91,500	W61930B1	82,350
1	1.85	EIPS	W62632A1	103,400	W61932B1	93,060
1-1/16	2.02	EIPS	W62634A1	123,800	W61934B1	111,507
1-1/8	2.34	EIPS	W62636A1	130,000	W61936B1	117,000
1-1/4	2.89	EIPS	W62640A1	159,800	W61940B1	143,200
1-3/8	3.5	EIPS	W62644A1	192,000	W61944B1	172,800
1-1/2	4.16	EIPS	W62648A1	228,000	W61948B1	205,200
1-5/8	4.88	EIPS	W62652A1	264,000	W61952B1	237,600
1-3/4	5.67	EIPS	W62656A1	306,000	W61956B1	275,400

Wesco Industries Ltd.
PowerStrand Wire Rope



7 x 7 & 7 x 19 Class Wire Rope "Aircraft Cable"

Galvanized 6 Outer Strands, Rotation Wire Rope

A commercial grade aircraft cable that makes for a very good all round small diameter rope. It is suitable for many applications including marine ropes, stage rigging, winch ropes, pulling cables, assemblies, seismic bracing and control Cables.

7x7 (6+1) WSC, Load bearing outer wires 42
Fill factor 0.55
7x19 (12+6+1) WSC, Load bearing outer wires 114
Fill factor 0.57
Manufactured to US Federal Specification RR-W-410F
Use of swivel is not permitted.

Diameter Inches	Weight per Foot Pounds	Grade Tensile	7 x 7 Construction		7 x 19 Construction	
			Wesco Stock Code	Minimum Strength Pounds	Wesco Stock Code	Minimum Strength Pounds
3/64	0.01	IPS	W70701B5C	270	-	-
1/16	0.01	IPS	W70702B5C	480	-	-
3/32	0.02	IPS	W70703B5C	920	-	-
1/8	0.03	IPS	W70704B5C	1,700	W71904B5C	2,000
5/32	0.05	IPS	-	-	W71905B5C	2,800
3/16	0.07	IPS	-	-	W71906B5C	4,200
7/32	0.09	IPS	-	-	W71907B5C	5,600
1/4	0.11	IPS	-	-	W71908B5C	7,000
5/16	0.17	IPS	-	-	W71910B5C	9,800
3/8	0.24	IPS	-	-	W71912B5C	14,400
1/2	0.41	IPS	-	-	W71916B5C	24,000
5/8	0.65	IPS	-	-	W71920B5C	37,000

Note: Not to be used on Aircraft

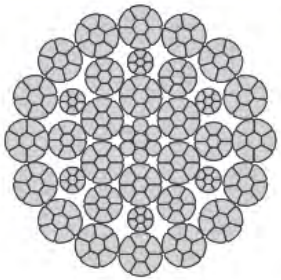
Example of Wire Rope



Overhead hoist – Standard cable is permitted since there is no “free end” to swivel with the load.



Example of a crane which requires a cable that is designed to prevent “free end” swivel when under load



PRO34K Compacted Wire Rope

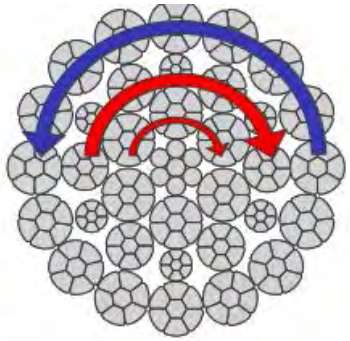
16 Outer Strands, Non-Rotating Wire Rope

An excellent 3 layer non rotating wire rope, class 35x7, with 16 outer compacted langs lay strands. This rope has a high tensile strength, a safe construction and the best in rotation resistance without compromising on flexibility and good resistance to drum crushing. Very low torque factor for big height hoists and lifts, with strong reliability combined with outstanding load capacity creates a best seller for telescopic mobile cranes, tower cranes, offshore cranes and any application where a non rotating rope is required.

34x7 (6+1) Load bearing outer wires 112

Fill factor 0.715 Torque factor 0.009

Use of swivel is permitted.



Diameter	Weight per Foot Pounds	Finish	Grade Tensile	Wesco Stock Code	Minimum Strength Pounds	Minimum Strength kN
8mm	0.19	Galvanized	2160 N/mm2	M080PRO34KB221R	13,937	62.0
9mm	0.27	Galvanized	2160 N/mm2	M090PRO34KB221R	17,759	79.0
3/8"	0.29	Galvanized	2160 N/mm2	M095PRO34KB221R	19,715	87.7
10mm	0.30	Galvanized	2160 N/mm2	M100PRO34KB221R	21,805	97.0
11mm	0.40	Galvanized	2160 N/mm2	M110PRO34KB221R	26,526	118.0
12mm	0.46	Galvanized	2160 N/mm2	M120PRO34KB221R	31,472	140.0
1/2"	0.50	Galvanized	2160 N/mm2	M127PRO34KB221R	35,068	156.0
13mm	0.54	Galvanized	2160 N/mm2	M130PRO34KB221R	36,417	162.0
14mm	0.64	Galvanized	2160 N/mm2	M140PRO34KB221R	42,937	191.0
15mm	0.73	Galvanized	2160 N/mm2	M150PRO34KB221R	49,231	219.0
16mm	0.85	Galvanized	2160 N/mm2	M160PRO34KB221R	55,975	249.0
17mm	0.91	Galvanized	2160 N/mm2	M170PRO34KB221R	62,944	280.0
18mm	1.06	Bright	2160 N/mm2	M180PRO34KB221*	70,812	315.0
19mm	1.18	Galvanized	2160 N/mm2	M190PRO34KB221*	78,905	351.0
20mm	1.30	Galvanized	2160 N/mm2	M200PRO34KB221R	87,447	389.0

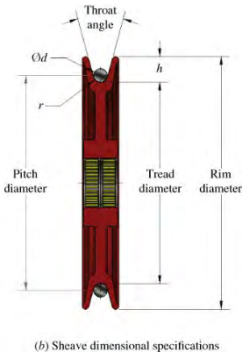


Wire Rope (Shigley)

- Stress in a wire rope is often dominated by the bending stress as the wire passes over a sheave, or drum.
- Note that the *wire rope stiffness*, E_r is <u>not</u> the same as the material strength of the individual wire.

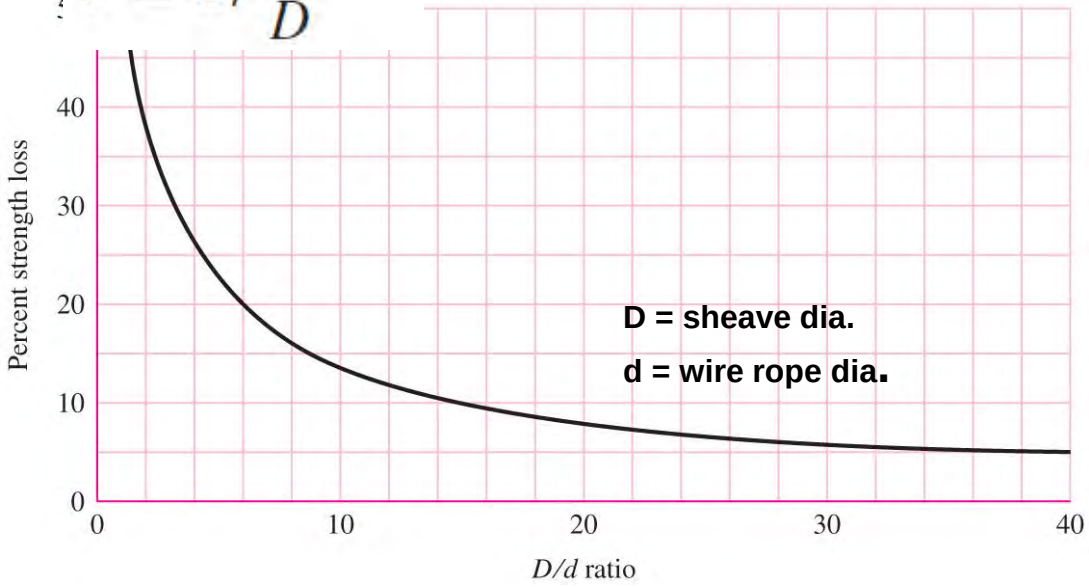


(*a*) Sheave with roller bearing



(*b*) Sheave dimensional specifications

$$\sigma = E_r \frac{d_w}{D}$$



Wire Rope

- Full bending occurs whenever the angle of departure is greater than about 12°
- Less than full bending occurs when less than one rope lay is in contact with the sheave.



- This normally occurs at angles of 10 or less. In this case the sheaves are often referred to as “knuckles.”
- The use of knuckle sheaves extend the overall diameter of the turn.
- Knuckles also reduce inertial impacts making them ideal for chairlifts and gondolas

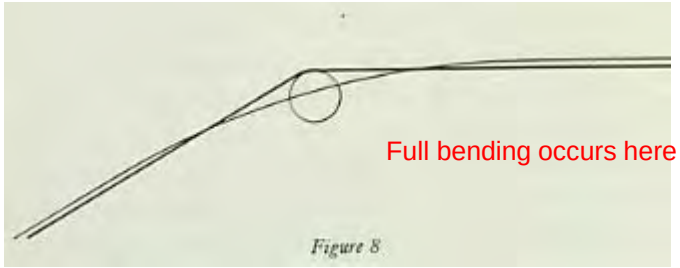
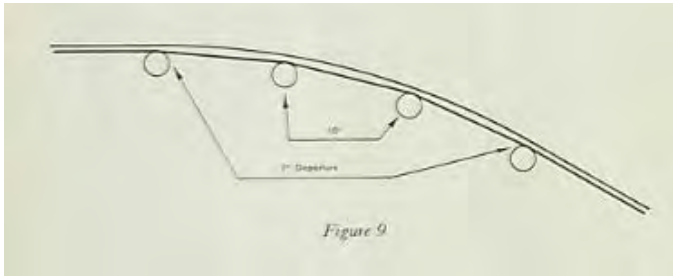
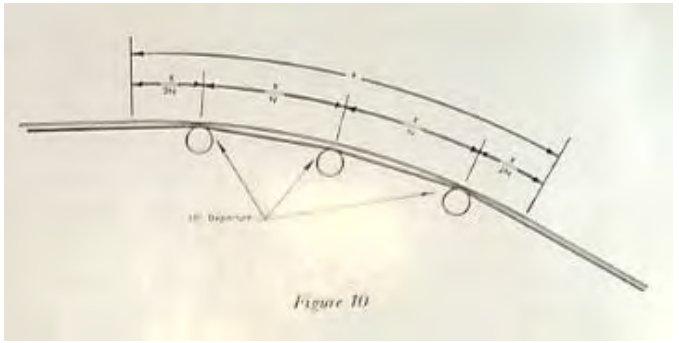


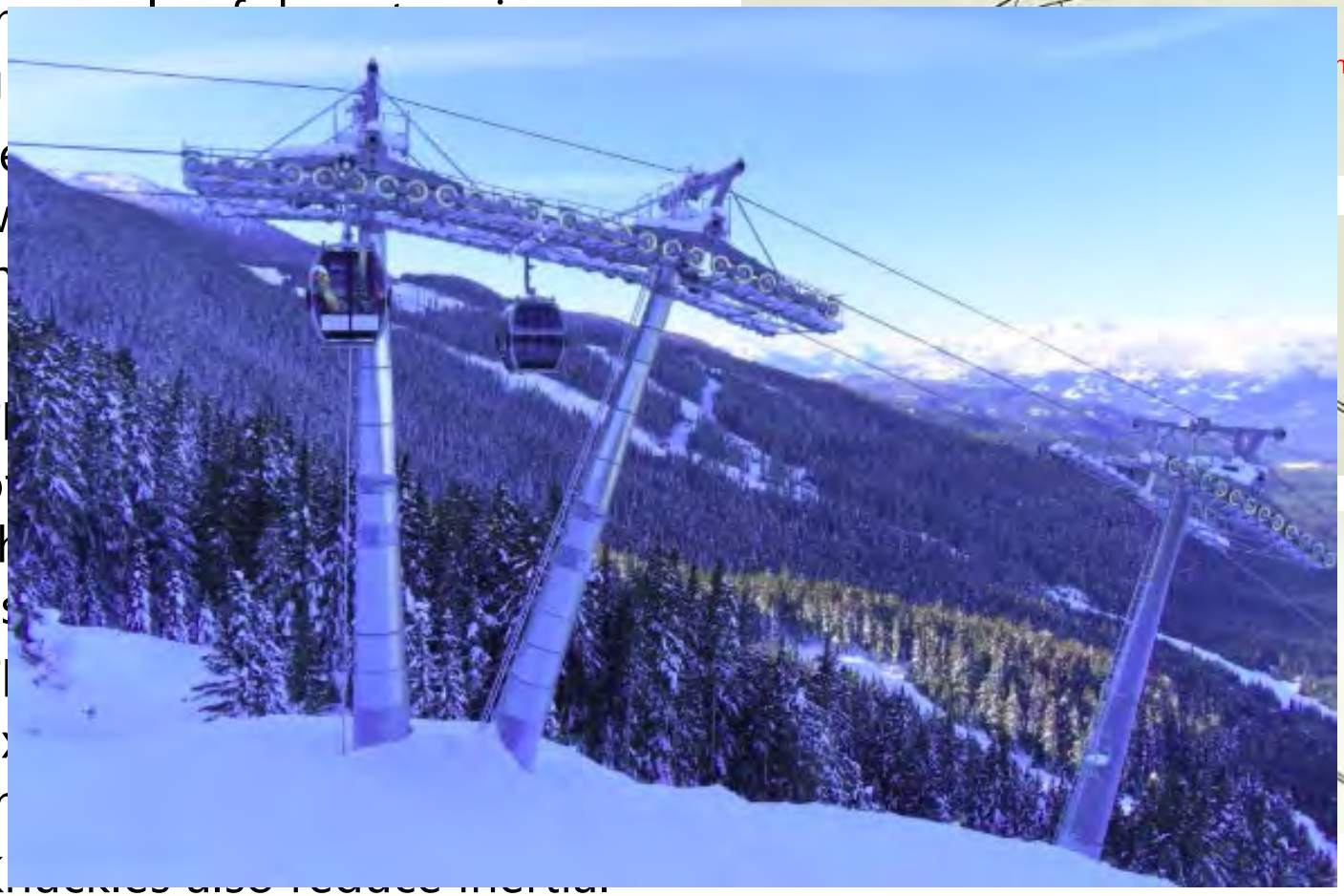
Figure 8


Figure 9


Figure 10

Wire Rope

- Full bending occurs whenever th
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  impacts making them ideal for chairlifts and gondolas



Wire Rope

E_{rope} Nominal S_u

Rope	Weight per Foot, lbf	Minimum Sheave Diameter, in	Standard Sizes d, in	Material	Size of Outer Wires	Modulus of Elasticity,* Mpsi	Strength,† kpsi
6 × 7 haulage	$1.50d^2$	$42d$	$\frac{1}{4}$ to $1\frac{1}{2}$	Monitor steel (Improved Plow Steel) Plow steel Mild plow steel	$d/9$ $d/9$ $d/9$	14 14 14	100 88 76
6 × 19 standard hoisting	$1.60d^2$	$26d$ to $34d$	$\frac{1}{4}$ to $2\frac{3}{4}$	Monitor steel Plow steel Mild plow steel	$d/13$ to $d/16$ $d/13$ to $d/16$ $d/13$ to $d/16$	12 12 12	106 93 80
6 × 37 special flexible	$1.55d^2$	$18d$	$\frac{1}{4}$ to $3\frac{1}{2}$	Monitor steel Plow steel	$d/22$ $d/22$	11 11	100 88
8 × 19 extra flexible	$1.45d^2$	$21d$ to $26d$	$\frac{1}{4}$ to $1\frac{1}{2}$	Monitor steel Plow steel	$d/15$ to $d/19$ $d/15$ to $d/19$	10 10	92 80
7 × 7 aircraft	$1.70d^2$	—	$\frac{1}{16}$ to $\frac{3}{8}$	Corrosion-resistant steel Carbon steel	— —	— —	124 124
7 × 9 aircraft	$1.75d^2$	—	$\frac{1}{8}$ to $1\frac{3}{8}$	Corrosion-resistant steel Carbon steel	— —	— —	135 143
19-wire aircraft	$2.15d^2$	—	$\frac{1}{32}$ to $\frac{5}{16}$	Corrosion-resistant steel Carbon steel	— —	— —	165 165

*The modulus of elasticity is only approximate; it is affected by the loads on the rope and, in general, increases with the life of the rope.

†The strength is based on the nominal area of the rope. The figures given are only approximate and are based on 1-in rope sizes and $\frac{1}{4}$-in aircraft-cable sizes.

Source: Compiled from *American Steel and Wire Company Handbook.*

Table 17–24

Equivalent bending load F_b

$$F_b = \sigma A_m = \frac{E_r d_w A_m}{D}$$

Wire Rope

Table 17–27

Some Useful Properties of 6 × 7, 6 × 19, and 6 × 37 Wire Ropes

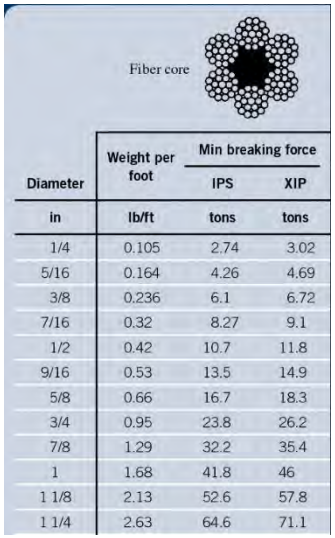
Wire Rope	Weight per Foot w, lbf/ft	Weight per Foot Including Core w, lbf/ft	Minimum Sheave Diameter D, in	Better Sheave Diameter D, in	Diameter of Wires d_w, in	Area of Metal A_m, in^2	Rope Young's Modulus E_r, psi
						A_m	E_r
6 × 7	$1.50d^2$		$42d$	$72d$	$0.111d$	$0.38d^2$	13×10^6
6 × 19	$1.60d^2$	$1.76d^2$	$30d$	$45d$	$0.067d$	$0.40d^2$	12×10^6
6 × 37	$1.55d^2$	$1.71d^2$	$18d$	$27d$	$0.048d$	$0.40d^2$	12×10^6

$$F_b = \sigma A_m = \frac{E_r d_w A_m}{D}$$

- The equivalent bending load F_b is defined as the wire tension giving the same tensile stress caused by sheave bending.
- It can be considered as an *effective* force

Wire Rope (Shigley)

- The ultimate strength of a wire rope will be function of the quality of steel used for the wires
- The estimate for the breaking force, F_u, of the wire rope is the nominal ultimate strength (Table 17-24) of wire applied over the nominal wire rope total area



Diameter	Weight per foot	Min breaking force	
		IPS	XIP
in	lb/ft	tons	tons
1/4	0.105	2.74	3.02
5/16	0.164	4.26	4.69
3/8	0.236	6.1	6.72
7/16	0.32	8.27	9.1
1/2	0.42	10.7	11.8
9/16	0.53	13.5	14.9
5/8	0.66	16.7	18.3
3/4	0.95	23.8	26.2
7/8	1.29	32.2	35.4
1	1.68	41.8	46
1 1/8	2.13	52.6	57.8
1 1/4	2.63	64.6	71.1

$$F_u = (S_{u\,nom})\,A = (S_{u\,nom})\,(\pi\, d^2/4)$$

For Example:

1" diameter Improved Plow Steel (IPS) 7 by 19 wire

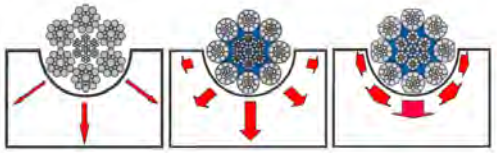
Nominal $S_{u\,min}$ = 106,000 psi (Table 17-24)

Ultimate Force (Breaking Force) = $106{,}000(\pi * 1^2/4)$ = 83,252 lbs = 41.6 tons

Wire Rope Sheave Bearing Pressures

$$p = \frac{2F}{dD}$$

where F = tensile force on rope
d = rope diameter
D = sheave diameter



Maximum Allowable Bearing Pressures(in psi)

Rope	Sheave Material: Wood[a]	Cast Iron[b]	Cast Steel[c]	Chilled Cast Irons[d]	Manganese Steel[e]
Regular lay:					
6 × 7	150	300	550	650	1470
6 × 19	250	480	900	1100	2400
6 × 37	300	585	1075	1325	3000
8 × 19	350	680	1260	1550	3500
Lang lay:					
6 × 7	165	350	600	715	1650
6 × 19	275	550	1000	1210	2750
6 × 37	330	660	1180	1450	3300

[a] On end grain of beech, hickory, or gum.

[b] For H_B (min.) = 125.

[c] 30–40 carbon; H_B (min.) = 160.

[d] Use only with uniform surface hardness.

[e] For high speeds with balanced sheaves having ground surfaces.

Wire Rope Safety Factors

- Factor of safety is defined as:

$$n = \frac{F_u}{F_t}$$

- F_t is the largest working tension based on static loads, shock loads, inertial loads, and sheave bearing loads.

Track cables	3.2
Guys	3.5
Mine shafts, ft:	
Up to 500	8.0
1000–2000	7.0
2000–3000	6.0
Over 3000	5.0
Hoisting	5.0
Haulage	6.0
Cranes and derricks	6.0
Electric hoists	7.0
Hand elevators	5.0
Private elevators	7.5
Hand dumbwaiter	4.5
Grain elevators	7.5

Passenger elevators, ft/min:	
50	7.60
300	9.20
800	11.25
1200	11.80
1500	11.90
Freight elevators, ft/min:	
50	6.65
300	8.20
800	10.00
1200	10.50
1500	10.55
Powered dumbwaiters, ft/min:	
50	4.8
300	6.6
500	8.0

Loading on Wire Rope (Shigley)

- The loading on the wire rope is estimated as the weight of the load and the weight of the fully extended steel rope
- This value is increased to account for dynamic loads due acceleration/deceleration of the load and wire. It is often given as the fraction of g force. This is defined as the tension force F_t
- The safety factor is normally F_u / F_t but Shigley recommends the removal of equivalent bending force, F_b from the ultimate force to give the *factor of safety for static loading*

$$n_s = \frac{F_u - F_b}{F_t} \qquad \textbf{(17 - 46)}$$

$$F_t = \left(\frac{W}{m} + wl\right)\left(1 + \frac{a}{g}\right)$$

where W = weight at the end of the rope (cage and load), lbf
m = number of wire ropes supporting the load
w = weight/foot of the wire rope, lbf/ft
l = maximum suspended length of rope, ft
a = maximum acceleration/deceleration experienced, ft/s^2
g = acceleration of gravity, ft/s^2 **= 32.2/ft/s^2**

Wire Rope Working Limits

SINGLE LEG WIRE ROPE ASSEMBLIES





Diameter Inches	Working Load Limit in Pounds		
	Vertical	Choker	Basket
1/4	1,300	960	2,600
5/16	2,000	1,480	4,000
3/8	2,800	2,200	5,600
1/2	5,000	3,800	10,200
5/8	7,800	5,800	15,600
3/4	11,200	8,200	22,600
7/8	15,200	11,200	30,000
1	19,800	14,400	40,000
1 1/8	24,000	18,200	48,000
1 1/4	30,000	22,000	60,000
1 3/8	36,000	26,000	72,000
1 1/2	42,000	32,000	84,000
1 5/8	48,000	36,000	96,000
1 3/4	56,000	42,000	114,000
1 7/8	64,000	48,000	126,000
2	74,000	56,000	146,000

- Rated Capacities basket hitch based on D/d ratio of 25.
- Rated Capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated Capacities based on design factor of 5:1.
- Horizontal Sling angles less than 30 degrees shall not be used.

WARNING: DO NOT EXCEED WORKING LOAD LIMIT

POWERSTRAND 2 LEG BRIDLE SLINGS



Diameter Inches	WLL 2 Leg Bridle in Pounds		
	60° to Load	45° to Load	30° to Load
1/4	2,200	1,820	1,300
5/16	3,400	2,800	2,000
3/8	5,000	4,000	2,800
1/2	8,800	7,200	5,000
5/8	13,800	11,000	7,800
3/4	19,400	15,800	11,200
7/8	26,000	22,000	15,200
1	34,000	28,000	19,600
1 1/8	42,000	34,000	24,000
1 1/4	52,000	42,000	30,000
1 3/8	62,000	50,000	36,000
1 1/2	74,000	60,000	42,000
1 5/8	84,000	70,000	48,000
1 3/4	98,000	80,000	56,000
1 7/8	112,000	92,000	64,000
2	126,000	104,000	74,000

- Rated Capacities basket hitch based on D/d ratio of 25.
- Rated Capacities based on pin diameter no larger than natural eye width or less than the nominal sling diameter.
- Rated Capacities based on design factor of 5:1.
- Horizontal Sling angles less than 30 degrees shall not be used.

WARNING: DO NOT EXCEED WORKING LOAD LIMIT

Wire Rope Fatigue Loading

- An S-N style plot is developed for wire ropes

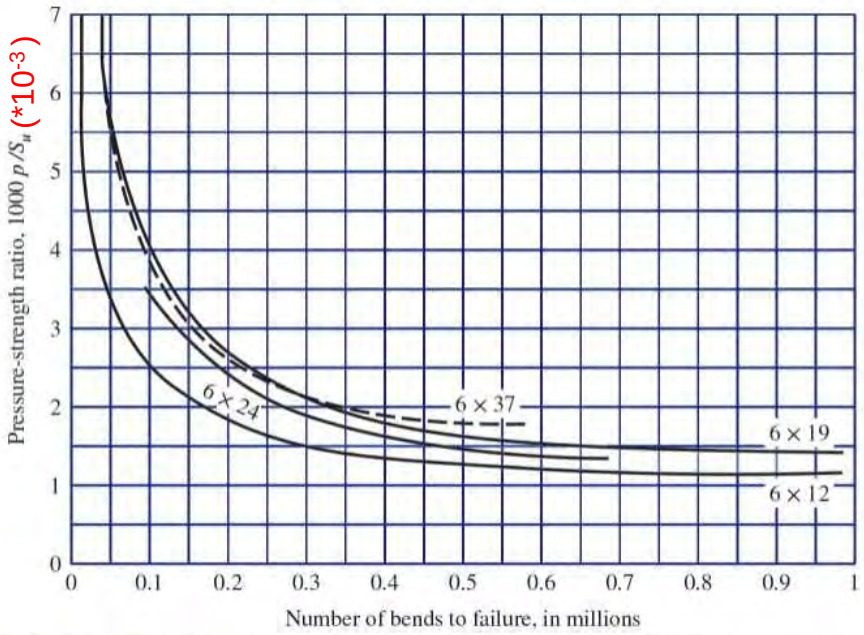


Figure 17–21 Experimentally determined relation between the fatigue life of wire rope and the sheave pressure.

The fatigue tensile strength in pounds for a specified life is

$$F_f = \frac{(p/S_u)S_u D d}{2}$$

where (p/S_u) = specified life, from Figure 17–21
S_u = ultimate tensile strength of the wires, psi
D = sheave or winch drum diameter, in
d = nominal wire rope size, in

Factor of safety for fatigue is

$$n_f = \frac{F_f - F_b}{F_t} \qquad \textbf{(17-45)}$$

Typical Strength of Individual Wires

Improved plow steel (monitor)	$240 < S_u < 280$ kpsi
Plow steel	$210 < S_u < 240$ kpsi
Mild plow steel	$180 < S_u < 210$ kpsi

CHOICE OF END FITTINGS

The selection of the right end fittings is as important as the right selection of wire rope for the application. The right selection and installation of these products is vital to maintain the assemblies integrity and strength. Rope fittings are generally subject to the same loadings as the wire rope they are used on, and should never be allowed to exceed the working load limit.



Selection of end fittings can be determined by the following:

A) Application.
B) Construction and lay of rope.
C) Safety Factor and Working Load Limit required.
D) Delivery required.
E) Site or Factory manufacture.

Rope fittings shall be properly designed for the service expected of them, and be installed and assembled to manufacturers specifications.

EFFICIENCY RATINGS

TERMINATIONS	EFFICIENCY RATING
WIRE ROPE SPLICE	
FLEMISH EYE SPLICE	1/4" TO 1" DIAMETER AND SMALLER 95%
	1 1/8" TO 2" DIAMETER 92.5%
	2 1/8" & LARGER DIAMETER 90%
PRESSED SLEEVE	1" DIAMETER AND SMALLER 95%
	1 1/8" TO 2" DIAMETER 92.5%
	2 1/8" & LARGER DIAMETER 90%
HAND TUCKED SPLICE	ONLY TO BE USED WHERE SLING ENDS ARE TERMINATED AND NOT ALLOWED TO ROTATE
	1/4" DIAMETER - 90%, 5/16" DIAMETER - 89%,
	3/8" DIAMETER - 88%, 7/16" DIAMETER - 87%,
	1/2" DIAMETER - 86%, 5/8" DIAMETER - 84%
	3/4" DIAMETER - 82%, 7/8" DIA. & GREATER - 80%
SWAGE SOCKET	OPEN OR CLOSED 100%
SPELTER SOCKET	OPEN OR CLOSED 100%
WEDGE SOCKET	75-90% DEPENDING ON DESIGN (Check with Wesco)
CLIPS	80% IF INSTALLED CORRECTLY





SPLICED EYES

Flemish Eye Splice

The flemish eye splice is fabricated by opening or unlaying the rope body into two parts, one having three strands and the other having the remaining three strands and the core. The rope is unlayed to allow the loop or eye to be formed by looping one part in one direction and the other part in the other direction and laying the rope back together. The strands are rolled back around the rope body. A metal sleeve is then slipped over the ends of the splice and pressed (swaged) to secure the ends to the body of the sling.



Care should be taken not to deform or damage the sleeves on these slings. Slings which have sleeves made of a different grade or type metal than the rope body may experience accelerated deterioration due to an electrochemical reaction between the two metals. This is particularly evident in salt water or brackish conditions.



Fold Back Mechanical Splice

The fold back splice is fabricated by forming a loop at the end of the rope, sliding one or more sleeves over the short end of the loop eye and pressing these sleeves to secure the end of the rope to the sling body. Sleeves can be either steel, stainless steel, aluminum or copper. Steel sleeves are the most commonly used sleeve due to their durability. Copper and aluminum sleeves are less durable.



A drawback to this type of sling is that the lifting capacity of the sling depends completely upon the integrity of the pressed or swaged joint. Should the metal sleeve(s) fail, the entire eye will also fail.

Care should be taken not to deform or damage the sleeves on these slings. Slings which have sleeves made of a different grade or type metal than the rope body may experience accelerated deterioration due to an electrochemical reaction between the two metals. This is particularly evident in salt water or brackish conditions.

WIRE ROPE CLIPS

Wire rope clips are useful for field installations. Correct installation with the use of a torque wrench, the termination efficiency will be 80%. It is important that installation instructions are followed carefully and the table below outlining minimum clips to be used and torque to be applied is strictly adhered to.



Wire rope clip installation is as follows:
1. Turn back specified amount of rope (see table) from thimble or loop. Apply first clip one base width from dead end of rope. Apply U-bolt over dead end of wire rope - live end rests in saddle (Never saddle a dead horse!). Tighten nuts evenly, alternating from one nut to the other until reaching the recommended torque.
2. When two clips are required, apply the second clip as near the loop or thimble as possible. Tighten nuts evenly, alternating until reaching the recommended torque. When more than two clips are required, apply the second clip as near the loop or thimble as possible, turn nuts on second clip firmly, but do not tighten. Proceed to Step 3.
3. When three or more clips are required, space additional clips between first two - take up rope slack - tighten nuts on all clips, alternating from one nut to the other until reaching the recommended torque.
4. Apply an initial load equal to or greater than the loads expected in use. Inspect for proper spacing and retighten the nuts to the

Step 1



Step 2



Step 3



Rope Sizes Inches	Minimum number of Clips	Amount of Rope to Turn Back in Inches	Torque in Foot Pounds
1/8	2	3 1/4	4.5
3/16	2	3 3/4	7.5
1/4	2	4 3/4	15
5/16	2	5 1/4	30
3/8	2	6 1/2	45
7/16	2	7	65
1/2	3	11 1/2	65
9/16	3	12	95
5/8	3	12	95
3/4	4	18	130
7/8	4	19	225
1	5	26	225
1 1/8	6	34	225
1 1/4	7	44	360
1 3/8	7	44	360
1 1/2	8	54	360
1 5/8	8	58	430
1 3/4	8	61	590
2	8	71	750
2 1/4	8	73	750
2 1/2	9	84	750
2 3/4	10	100	750
3	10	106	1,200

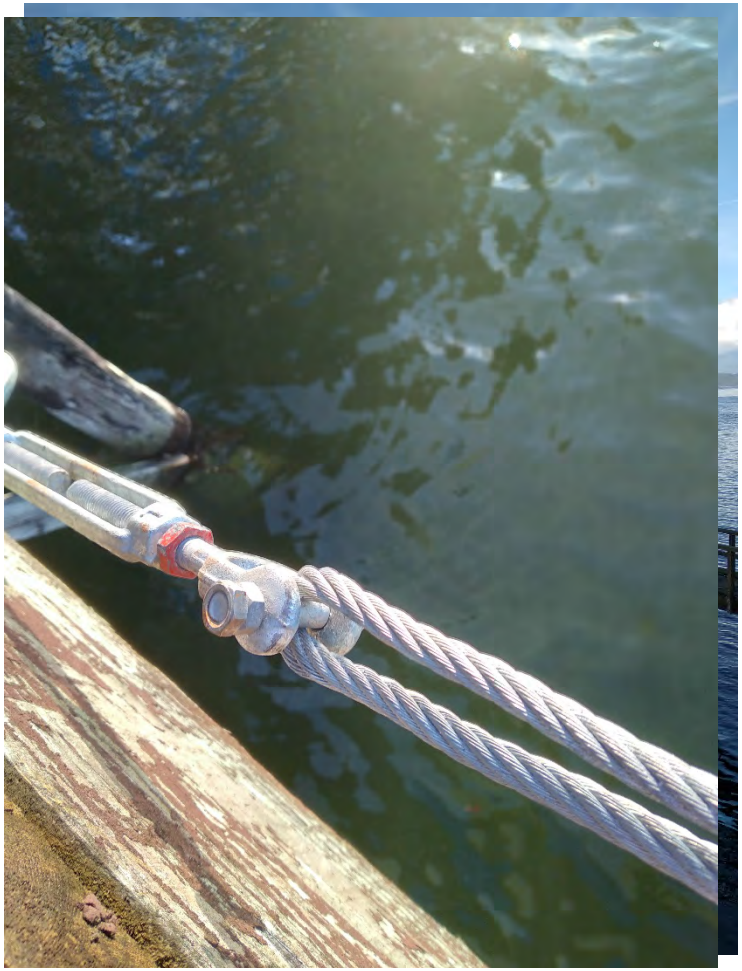






Lunch atop Rockefeller Center 1932

Wire Rope Example





Wire Rope Safety

- Wire ropes are normally lubricated to reduce friction, wear, and protect from corrosion
- Wire ropes must regularly be inspected for any signs of breakage, crushing, and corrosion
- Regulations are in place to govern the inspection and replacement of wire ropes









Wire Rope Safety

- Wire ropes must be properly reeled (spooled) to ensure not kinks occur in the rope.
- Figure 9 highlights how the integrity of the rope is compromised due to a kink being formed. Wire failure is common (Figure 11).



Figure 5



Figure 6



Figure 7



Figure 8

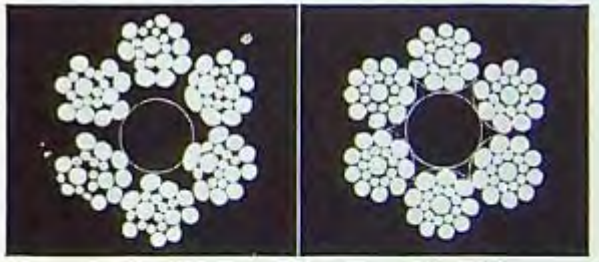


Figure 9 Figure 10



Figure 11—Failure Resulting from Kink Damage.